

#PROJECT ON PANDAS FUNCTONS

```
import pandas as pd

data = {
    "StudentID": range(301, 321),
    "Name": [
        "Alice Brown", "Ben Johnson", "Clara Davis", "Daniel Lee", "Eva Wilson",
        "Frank Miller", "Grace Taylor", "Henry Clark", "Irene White", "Jack Lewis",
        "Jaspreet singh", "Liam Scott", "Mia Adams", "Noah Carter", "Olivia Green",',
        "Paul Walker", "Quinn Rivera", "Ryan Hall", "Sophia King", "Thomas Allen"
    ],
    "Major": [
        "Computer Science", "Mechanical Eng.", "Psychology", "Business Mgmt.", "Biology",
        "Computer Science", "English Literature", "Mathematics", "Chemistry", "Business Mgmt.",
        "Biology", "Computer Science", "Mathematics", "Mechanical Eng.", "Psychology",
        "Business Mgmt.", "English Literature", "Biology", "Mathematics", "Computer Science"
    ],
    "GPA": [
        3.9, 3.3, 3.5, 3.2, 3.8, 2.9, 3.6, 3.7, 3.4, 3.1,
        3.9, 3.6, 3.2, 3.0, 3.8, 3.5, 3.7, 2.8, 3.9, 3.4
    ],
    "State": [
        "California", "Texas", "Florida", "Illinois", "New York", "California", "Ohio",
        "Texas", "Washington", "Florida", "California", "New York", "Texas", "Illinois",
        "Florida", "Ohio", "Texas", "California", "Washington", "Illinois"
    ]
}

df = pd.DataFrame(data)
```

df

	StudentID	Name	Major	GPA	State	
0	301	Alice Brown	Computer Science	3.9	California	
1	302	Ben Johnson	Mechanical Eng.	3.3	Texas	
2	303	Clara Davis	Psychology	3.5	Florida	
3	304	Daniel Lee	Business Mgmt.	3.2	Illinois	
4	305	Eva Wilson	Biology	3.8	New York	
5	306	Frank Miller	Computer Science	2.9	California	
6	307	Grace Taylor	English Literature	3.6	Ohio	
7	308	Henry Clark	Mathematics	3.7	Texas	
8	309	Irene White	Chemistry	3.4	Washington	
9	310	Jack Lewis	Business Mgmt.	3.1	Florida	
10	311	Jaspreet singh	Biology	3.9	California	
11	312	Liam Scott	Computer Science	3.6	New York	
12	313	Mia Adams	Mathematics	3.2	Texas	
13	314	Noah Carter	Mechanical Eng.	3.0	Illinois	
14	315	Olivia Green	Psychology	3.8	Florida	
15	316	Paul Walker	Business Mgmt.	3.5	Ohio	
16	317	Quinn Rivera	English Literature	3.7	Texas	
17	318	Ryan Hall	Biology	2.8	California	
18	319	Sophia King	Mathematics	3.9	Washington	
19	320	Thomas Allen	Computer Science	3.4	Illinois	

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
#Display the first 5 rows of the dataset to get an overview of the data

df.head(5)
```

	StudentID	Name	Major	GPA	State
0	301	Alice Brown	Computer Science	3.9	California
1	302	Ben Johnson	Mechanical Eng.	3.3	Texas
2	303	Clara Davis	Psychology	3.5	Florida
3	304	Daniel Lee	Business Mgmt.	3.2	Illinois
4	305	Eva Wilson	Biology	3.8	New York

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```
#What happens if you run df.tail(10)? How many rows are shown?  
  
df.tail(10)
```

	StudentID	Name	Major	GPA	State
10	311	Jaspreet singh	Biology	3.9	California
11	312	Liam Scott	Computer Science	3.6	New York
12	313	Mia Adams	Mathematics	3.2	Texas
13	314	Noah Carter	Mechanical Eng.	3.0	Illinois
14	315	Olivia Green	Psychology	3.8	Florida
15	316	Paul Walker	Business Mgmt.	3.5	Ohio
16	317	Quinn Rivera	English Literature	3.7	Texas
17	318	Ryan Hall	Biology	2.8	California
18	319	Sophia King	Mathematics	3.9	Washington
19	320	Thomas Allen	Computer Science	3.4	Illinois

```
#get a summary specifically for GPA  
  
df["GPA"].describe()
```

	GPA
count	20.000000
mean	3.460000
std	0.342437
min	2.800000
25%	3.200000
50%	3.500000
75%	3.725000
max	3.900000

dtype: float64

```
#check the total number of rows and columns in the dataset  
  
df.info()
```

```
<class 'pandas.core.frame.DataFrame'  
RangeIndex: 20 entries, 0 to 19  
Data columns (total 5 columns):  
#   Column      Non-Null Count  Dtype  
---  ---      -  
0   StudentID    20 non-null     int64  
1   Name         20 non-null     object  
2   Major        20 non-null     object  
3   GPA          20 non-null     float64  
4   State        20 non-null     object  
dtypes: float64(1), int64(1), object(3)  
memory usage: 932.0+ bytes
```

```
#Verify that your dataset has 20 students and 5 columns  
  
df.shape  
  
(20, 5)
```

```
#Selection & Indexing
#How do you select the Name and GPA columns from the DataFrame?
```

```
df[['Name', 'GPA']]
```

	Name	GPA
0	Alice Brown	3.9
1	Ben Johnson	3.3
2	Clara Davis	3.5
3	Daniel Lee	3.2
4	Eva Wilson	3.8
5	Frank Miller	2.9
6	Grace Taylor	3.6
7	Henry Clark	3.7
8	Irene White	3.4
9	Jack Lewis	3.1
10	Jaspreet singh	3.9
11	Liam Scott	3.6
12	Mia Adams	3.2
13	Noah Carter	3.0
14	Olivia Green	3.8
15	Paul Walker	3.5
16	Quinn Rivera	3.7
17	Ryan Hall	2.8
18	Sophia King	3.9
19	Thomas Allen	3.4

```
#How do you select the row for the student with StudentID 308?
```

```
df[df['StudentID'] == 308]
```

	StudentID	Name	Major	GPA	State
7	308	Henry Clark	Mathematics	3.7	Texas

```
#How do you select the first 3 rows and the first 2 columns?
```

```
df.iloc[:3, :2]
```

	StudentID	Name
0	301	Alice Brown
1	302	Ben Johnson
2	303	Clara Davis

```
#Filtering & Conditional Selection
```

```
#How do you find all students with a GPA greater than 3.5?
```

```
df[df['GPA'] > 3.5]
```

	StudentID	Name	Major	GPA	State	
0	301	Alice Brown	Computer Science	3.9	California	
4	305	Eva Wilson	Biology	3.8	New York	
6	307	Grace Taylor	English Literature	3.6	Ohio	
7	308	Henry Clark	Mathematics	3.7	Texas	
10	311	Jaspreet singh	Biology	3.9	California	
11	312	Liam Scott	Computer Science	3.6	New York	
14	315	Olivia Green	Psychology	3.8	Florida	
16	317	Quinn Rivera	English Literature	3.7	Texas	
18	319	Sophia King	Mathematics	3.9	Washington	

#How do you filter students whose Major is Computer Science?

```
df[df['Major'] == 'Computer Science']
```

	StudentID	Name	Major	GPA	State	
0	301	Alice Brown	Computer Science	3.9	California	
5	306	Frank Miller	Computer Science	2.9	California	
11	312	Liam Scott	Computer Science	3.6	New York	
19	320	Thomas Allen	Computer Science	3.4	Illinois	

#How do you select students from California with a GPA above 3.0?

```
df[(df['State'] == 'California') & (df['GPA'] > 3.0)]
```

	StudentID	Name	Major	GPA	State	
0	301	Alice Brown	Computer Science	3.9	California	
10	311	Jaspreetsingh	Biology	3.9	California	

#How do you find students whose name starts with “A” or “B”?

```
df[df['Name'].str.startswith(('A', 'B'))]
```

	StudentID	Name	Major	GPA	State	
0	301	Alice Brown	Computer Science	3.9	California	
1	302	Ben Johnson	Mechanical Eng.	3.3	Texas	

#Sorting

#How do you sort the DataFrame by GPA in descending order?

```
df.sort_values(by='GPA', ascending=False)
```

	StudentID	Name	Major	GPA	State	
0	301	Alice Brown	Computer Science	3.9	California	
10	311	Jaspreet singh	Biology	3.9	California	
18	319	Sophia King	Mathematics	3.9	Washington	
14	315	Olivia Green	Psychology	3.8	Florida	
4	305	Eva Wilson	Biology	3.8	New York	
7	308	Henry Clark	Mathematics	3.7	Texas	
16	317	Quinn Rivera	English Literature	3.7	Texas	
11	312	Liam Scott	Computer Science	3.6	New York	
6	307	Grace Taylor	English Literature	3.6	Ohio	
15	316	Paul Walker	Business Mgmt.	3.5	Ohio	
2	303	Clara Davis	Psychology	3.5	Florida	
19	320	Thomas Allen	Computer Science	3.4	Illinois	
8	309	Irene White	Chemistry	3.4	Washington	
1	302	Ben Johnson	Mechanical Eng.	3.3	Texas	
3	304	Daniel Lee	Business Mgmt.	3.2	Illinois	
12	313	Mia Adams	Mathematics	3.2	Texas	
9	310	Jack Lewis	Business Mgmt.	3.1	Florida	
13	314	Noah Carter	Mechanical Eng.	3.0	Illinois	
5	306	Frank Miller	Computer Science	2.9	California	
17	318	Ryan Hall	Biology	2.8	California	

#How do you sort students first by State and then by GPA?

```
df.sort_values(by=['State', 'GPA'])
```

	StudentID	Name	Major	GPA	State	
17	318	Ryan Hall	Biology	2.8	California	
5	306	Frank Miller	Computer Science	2.9	California	
0	301	Alice Brown	Computer Science	3.9	California	
10	311	Jaspreet singh	Biology	3.9	California	
9	310	Jack Lewis	Business Mgmt.	3.1	Florida	
2	303	Clara Davis	Psychology	3.5	Florida	
14	315	Olivia Green	Psychology	3.8	Florida	
13	314	Noah Carter	Mechanical Eng.	3.0	Illinois	
3	304	Daniel Lee	Business Mgmt.	3.2	Illinois	
19	320	Thomas Allen	Computer Science	3.4	Illinois	
11	312	Liam Scott	Computer Science	3.6	New York	
4	305	Eva Wilson	Biology	3.8	New York	
15	316	Paul Walker	Business Mgmt.	3.5	Ohio	
6	307	Grace Taylor	English Literature	3.6	Ohio	
12	313	Mia Adams	Mathematics	3.2	Texas	
1	302	Ben Johnson	Mechanical Eng.	3.3	Texas	
7	308	Henry Clark	Mathematics	3.7	Texas	
16	317	Quinn Rivera	English Literature	3.7	Texas	
8	309	Irene White	Chemistry	3.4	Washington	
18	319	Sophia King	Mathematics	3.9	Washington	

#Aggregation & Grouping

#how do you calculate the average GPA for the entire dataset?

```
df['GPA'].mean()
```

```
np.float64(3.46)
```

```
#How do you find the highest GPA for each Major?
```

```
df.groupby('Major')['GPA'].max()
```

GPA	
Major	
Biology	3.9
Business Mgmt.	3.5
Chemistry	3.4
Computer Science	3.9
English Literature	3.7
Mathematics	3.9
Mechanical Eng.	3.3
Psychology	3.8

```
dtype: float64
```

```
#How do you count the number of students in each State?
```

```
df['State'].value_counts()
```

count	
State	
California	4
Texas	4
Florida	3
Illinois	3
New York	2
Ohio	2
Washington	2

```
dtype: int64
```

```
#how do you calculate the average GPA for students grouped by Major?
```

```
df.groupby('Major')['GPA'].mean()
```

GPA	
Major	
Biology	3.500000
Business Mgmt.	3.266667
Chemistry	3.400000
Computer Science	3.450000
English Literature	3.650000
Mathematics	3.600000
Mechanical Eng.	3.150000
Psychology	3.650000

```
dtype: float64
```

```
#Adding & Modifying Columns
#How do you increase every GPA by 0.1 for all students?
```

```
df['GPA'] = df['GPA'] + 0.1
df
```

	StudentID	Name	Major	GPA	State	
0	301	Alice Brown	Computer Science	4.0	California	
1	302	Ben Johnson	Mechanical Eng.	3.4	Texas	
2	303	Clara Davis	Psychology	3.6	Florida	
3	304	Daniel Lee	Business Mgmt.	3.3	Illinois	
4	305	Eva Wilson	Biology	3.9	New York	
5	306	Frank Miller	Computer Science	3.0	California	
6	307	Grace Taylor	English Literature	3.7	Ohio	
7	308	Henry Clark	Mathematics	3.8	Texas	
8	309	Irene White	Chemistry	3.5	Washington	
9	310	Jack Lewis	Business Mgmt.	3.2	Florida	
10	311	Jaspreet singh	Biology	4.0	California	
11	312	Liam Scott	Computer Science	3.7	New York	
12	313	Mia Adams	Mathematics	3.3	Texas	
13	314	Noah Carter	Mechanical Eng.	3.1	Illinois	
14	315	Olivia Green	Psychology	3.9	Florida	
15	316	Paul Walker	Business Mgmt.	3.6	Ohio	
16	317	Quinn Rivera	English Literature	3.8	Texas	
17	318	Ryan Hall	Biology	2.9	California	
18	319	Sophia King	Mathematics	4.0	Washington	
19	320	Thomas Allen	Computer Science	3.5	Illinois	

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
#How do you create a new column Pass that is True if GPA ≥ 3.0 and False otherwise?

df['Pass'] = df['GPA'] >= 3.0
df
```

	StudentID	Name	Major	GPA	State	Pass	
0	301	Alice Brown	Computer Science	4.0	California	True	
1	302	Ben Johnson	Mechanical Eng.	3.4	Texas	True	
2	303	Clara Davis	Psychology	3.6	Florida	True	
3	304	Daniel Lee	Business Mgmt.	3.3	Illinois	True	
4	305	Eva Wilson	Biology	3.9	New York	True	
5	306	Frank Miller	Computer Science	3.0	California	True	
6	307	Grace Taylor	English Literature	3.7	Ohio	True	
7	308	Henry Clark	Mathematics	3.8	Texas	True	
8	309	Irene White	Chemistry	3.5	Washington	True	
9	310	Jack Lewis	Business Mgmt.	3.2	Florida	True	
10	311	Jaspreet singh	Biology	4.0	California	True	
11	312	Liam Scott	Computer Science	3.7	New York	True	
12	313	Mia Adams	Mathematics	3.3	Texas	True	
13	314	Noah Carter	Mechanical Eng.	3.1	Illinois	True	
14	315	Olivia Green	Psychology	3.9	Florida	True	
15	316	Paul Walker	Business Mgmt.	3.6	Ohio	True	
16	317	Quinn Rivera	English Literature	3.8	Texas	True	
17	318	Ryan Hall	Biology	2.9	California	False	
18	319	Sophia King	Mathematics	4.0	Washington	True	
19	320	Thomas Allen	Computer Science	3.5	Illinois	True	

Next steps:

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#How do you create a new column Honor that shows "Yes" if GPA $\geq$ 3.7, otherwise "No"?

```
df['Honor'] = df['GPA'].apply(lambda x: 'Yes' if x >= 3.7 else 'No')
df
```

	StudentID	Name	Major	GPA	State	Pass	Honor	
0	301	Alice Brown	Computer Science	4.0	California	True	Yes	
1	302	Ben Johnson	Mechanical Eng.	3.4	Texas	True	No	
2	303	Clara Davis	Psychology	3.6	Florida	True	No	
3	304	Daniel Lee	Business Mgmt.	3.3	Illinois	True	No	
4	305	Eva Wilson	Biology	3.9	New York	True	Yes	
5	306	Frank Miller	Computer Science	3.0	California	True	No	
6	307	Grace Taylor	English Literature	3.7	Ohio	True	Yes	
7	308	Henry Clark	Mathematics	3.8	Texas	True	Yes	
8	309	Irene White	Chemistry	3.5	Washington	True	No	
9	310	Jack Lewis	Business Mgmt.	3.2	Florida	True	No	
10	311	Jaspreet singh	Biology	4.0	California	True	Yes	
11	312	Liam Scott	Computer Science	3.7	New York	True	Yes	
12	313	Mia Adams	Mathematics	3.3	Texas	True	No	
13	314	Noah Carter	Mechanical Eng.	3.1	Illinois	True	No	
14	315	Olivia Green	Psychology	3.9	Florida	True	Yes	
15	316	Paul Walker	Business Mgmt.	3.6	Ohio	True	No	
16	317	Quinn Rivera	English Literature	3.8	Texas	True	Yes	
17	318	Ryan Hall	Biology	2.9	California	False	No	
18	319	Sophia King	Mathematics	4.0	Washington	True	Yes	
19	320	Thomas Allen	Computer Science	3.5	Illinois	True	No	

Next steps:

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```
#String Operations
#How do you extract the first names from the Name column?

df['First_Name'] = df['Name'].str.split().str[0]
df
```

	StudentID	Name	Major	GPA	State	Pass	Honor	First_Name	
0	301	Alice Brown	Computer Science	4.0	California	True	Yes	Alice	
1	302	Ben Johnson	Mechanical Eng.	3.4	Texas	True	No	Ben	
2	303	Clara Davis	Psychology	3.6	Florida	True	No	Clara	
3	304	Daniel Lee	Business Mgmt.	3.3	Illinois	True	No	Daniel	
4	305	Eva Wilson	Biology	3.9	New York	True	Yes	Eva	
5	306	Frank Miller	Computer Science	3.0	California	True	No	Frank	
6	307	Grace Taylor	English Literature	3.7	Ohio	True	Yes	Grace	
7	308	Henry Clark	Mathematics	3.8	Texas	True	Yes	Henry	
8	309	Irene White	Chemistry	3.5	Washington	True	No	Irene	
9	310	Jack Lewis	Business Mgmt.	3.2	Florida	True	No	Jack	
10	311	Jaspreet singh	Biology	4.0	California	True	Yes	Karen	
11	312	Liam Scott	Computer Science	3.7	New York	True	Yes	Liam	
12	313	Mia Adams	Mathematics	3.3	Texas	True	No	Mia	
13	314	Noah Carter	Mechanical Eng.	3.1	Illinois	True	No	Noah	
14	315	Olivia Green	Psychology	3.9	Florida	True	Yes	Olivia	
15	316	Paul Walker	Business Mgmt.	3.6	Ohio	True	No	Paul	
16	317	Quinn Rivera	English Literature	3.8	Texas	True	Yes	Quinn	
17	318	Ryan Hall	Biology	2.9	California	False	No	Ryan	
18	319	Sophia King	Mathematics	4.0	Washington	True	Yes	Sophia	

```
import pandas as pd

data = {
    "StudentID": range(301, 321),
    "Name": [
        "Alice Brown", "Ben Johnson", "Clara Davis", "Daniel Lee", "Eva Wilson",
        "Frank Miller", "Grace Taylor", "Henry Clark", "Irene White", "Jack Lewis",
        "Jaspreet singh", "Liam Scott", "Mia Adams", "Noah Carter", "Olivia Green", ' ',
        "Paul Walker", "Quinn Rivera", "Ryan Hall", "Sophia King", "Thomas Allen"
    ],
    "Major": [
        "Computer Science", "Mechanical Eng.", "Psychology", "Business Mgmt.", "Biology",
        "Computer Science", "English Literature", "Mathematics", "Chemistry", "Business Mgmt.",
        "Biology", "Computer Science", "Mathematics", "Mechanical Eng.", "Psychology",
        "Business Mgmt.", "English Literature", "Biology", "Mathematics", "Computer Science"
    ],
    "GPA": [
        3.9, 3.3, 3.5, 3.2, 3.8, 2.9, 3.6, 3.7, 3.4, 3.1,
        3.9, 3.6, 3.2, 3.0, 3.8, 3.5, 3.7, 2.8, 3.9, 3.4
    ],
    "State": [
        "California", "Texas", "Florida", "Illinois", "New York", "California", "Ohio",
        "Texas", "Washington", "Florida", "California", "New York", "Texas", "Illinois",
        "Florida", "Ohio", "Texas", "California", "Washington", "Illinois"
    ]
}

df = pd.DataFrame(data)
```

df

	StudentID	Name	Major	GPA	State	
0	301	Alice Brown	Computer Science	3.9	California	
1	302	Ben Johnson	Mechanical Eng.	3.3	Texas	
2	303	Clara Davis	Psychology	3.5	Florida	
3	304	Daniel Lee	Business Mgmt.	3.2	Illinois	
4	305	Eva Wilson	Biology	3.8	New York	
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6	307	Grace Taylor	English Literature	3.6	Ohio	
7	308	Henry Clark	Mathematics	3.7	Texas	
8	309	Irene White	Chemistry	3.4	Washington	
9	310	Jack Lewis	Business Mgmt.	3.1	Florida	
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11	312	Liam Scott	Computer Science	3.6	New York	
12	313	Mia Adams	Mathematics	3.2	Texas	
13	314	Noah Carter	Mechanical Eng.	3.0	Illinois	
14	315	Olivia Green	Psychology	3.8	Florida	
15	316	Paul Walker	Business Mgmt.	3.5	Ohio	
16	317	Quinn Rivera	English Literature	3.7	Texas	
17	318	Ryan Hall	Biology	2.8	California	
18	319	Sophia King	Mathematics	3.9	Washington	
19	320	Thomas Allen	Computer Science	3.4	Illinois	

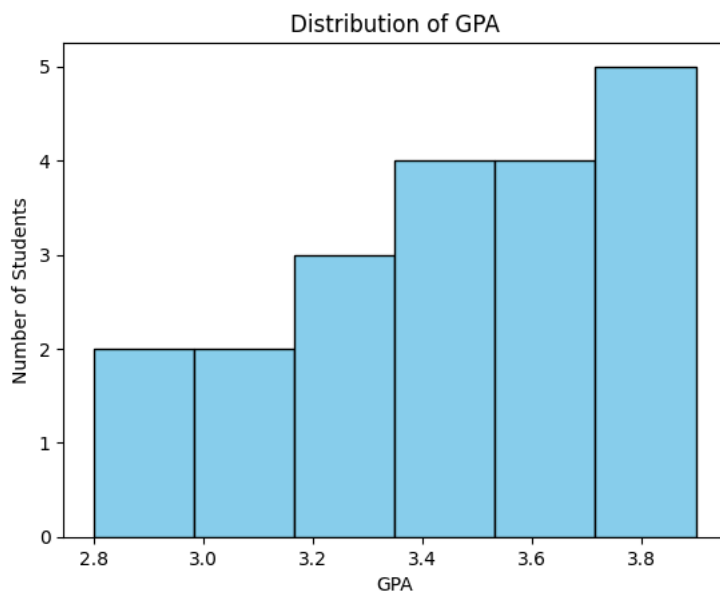
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```
#GRAPHS
#HISTOGRAM GRAPH
#Plot a histogram of the students' GPA.

import matplotlib.pyplot as plt

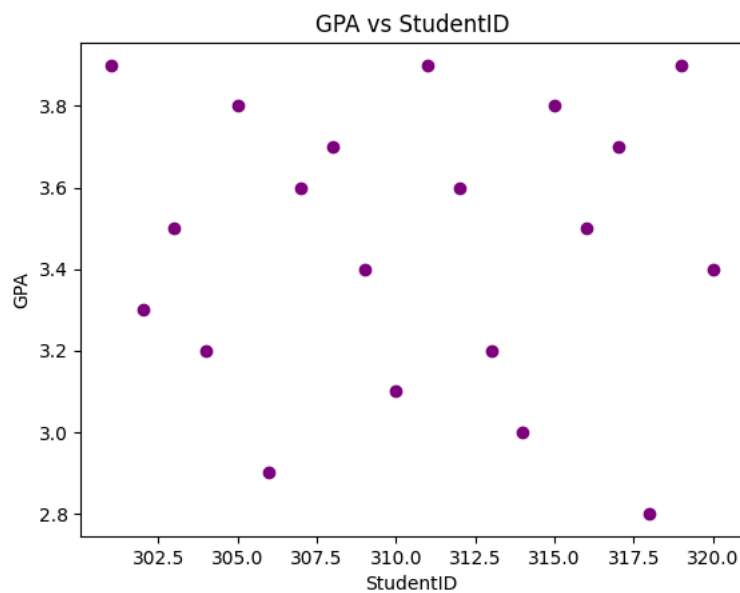
plt.hist(df['GPA'], bins=6, color='skyblue', edgecolor='black')
plt.title('Distribution of GPA')
plt.xlabel('GPA')
```

```
plt.ylabel('Number of Students')  
plt.show()
```



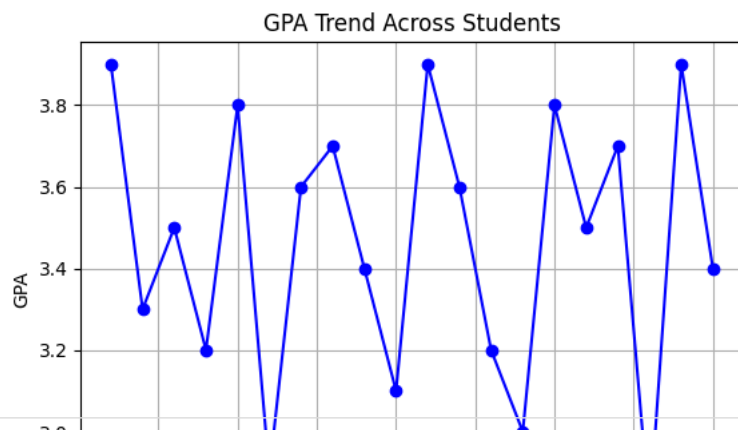
```
#SCATTER GRAPH  
#Plot GPA vs StudentID to see individual performance trends.
```

```
plt.scatter(df['StudentID'], df['GPA'], color='purple')  
plt.title('GPA vs StudentID')  
plt.xlabel('StudentID')  
plt.ylabel('GPA')  
plt.show()
```



```
#LINE GRAPH  
#Plot GPA against StudentID as a line plot to see GPA trends across students.
```

```
import matplotlib.pyplot as plt  
  
plt.plot(df['StudentID'], df['GPA'], marker='o', color='blue', linestyle='--')  
plt.title('GPA Trend Across Students')  
plt.xlabel('StudentID')  
plt.ylabel('GPA')  
plt.grid(True)  
plt.show()
```



```
#BAR GRAPH
# Count students per Major

major_counts = df['Major'].value_counts()

# Plot bar graph
major_counts.plot(kind='bar', color='skyblue', edgecolor='black')
plt.title('Number of Students per Major')
plt.xlabel('Major')
plt.ylabel('Number of Students')
plt.xticks(rotation=45, ha='right')
plt.show()
```

